Let

$$\begin{array}{ccc} A & \xrightarrow{i} & A \\ & {}_k\nwarrow \quad \swarrow_j & \\ & B & \end{array} \qquad \partial = jk : B \to B$$

be an exact couple. We form the exact couple

$$\begin{array}{ccc} i(A)=A' & \xrightarrow{i'} & A' \\ & {}_{k'}\nwarrow \quad \swarrow_{j'} & \\ & H(B)=\ker\partial/\operatorname{im}\partial & \end{array}$$

$k[b]=[kb]$, $j'(ia)=[ja]$

We need to check everything is well-defined and exact:

- ja is a cycle: $\partial ja = jkja = 0$
- j is well-defined: $i(a-a')=0 \Rightarrow a-a'=kb \Rightarrow j(a-a')=jkb=\partial b$
- k is well-defined: $[b]=[b'] \Rightarrow b-b'=\partial\tilde b \Rightarrow kb-kb'=k\partial\tilde b = kjk\tilde b = 0$

Now let (K,D) be a filtered complex, i.e.

$$\dots = K = K = K_0 \supset K_1 \supset K_2 \supset \dots$$

with $D(K_i)\subset K_i$ and $D^2=0$. Let $GK=\bigoplus K_p/K_{p+1}$ denote the associated graded. Let

$A = \dots \oplus K \oplus K \oplus K_1 \oplus K_2 \oplus \dots$

$A = \dots \oplus K \oplus K \oplus K_1 \oplus K_2 \oplus \dots$

Note that $B = A/i(A) = GK$. We get a s.e.s. of complexes

$$0 \to A \xrightarrow{i} A \xrightarrow{j} B \to 0,$$

and hence a l.e.s.

$$\dots \to H^k(A) \xrightarrow{i} H^k(A) \xrightarrow{j_*} H^k(B) \xrightarrow{k} H^{k+1}(A) \to \dots$$

i.e.

$$\begin{array}{ccc} H(A) & \xrightarrow{i} & H(A) \\ & {}_{k}\nwarrow \quad \swarrow_{j_*} & \\ & H(B) & \end{array} \quad = \quad \begin{array}{ccc} A_1 & \xrightarrow{i} & A_1 \\ & {}_{k_1}\nwarrow \quad \swarrow_{j_1} & \\ & B_1 & \end{array}$$

By the first part, this induces a sequence

$$\begin{array}{ccc} A_r & \to & A_r \\ & \nwarrow \quad \swarrow & \\ & B_r & \end{array}$$

Example: Let

$A = \dots \oplus K_0 \oplus K_1 \oplus K_2 \oplus K_3$,

$B = K/K_1 \oplus K_1/K_2 \oplus K_2/K_3 \oplus K_3$.

Note: Actually, $A_1 = H(K)\oplus H(K)\oplus H(K_1)\oplus H(K_2)$, $A_2 = i(A_1)$, etc. We included the maps i etc. to emphasise them becoming inclusions.

Then

$A_1 = H(K) \xleftarrow{i} H(K_1) \xleftarrow{i} H(K_2) \xleftarrow{i} H(K_3) \leftarrow 0,$

$A_2 = \operatorname{Im}(i: A_1 \to A_1)$

$= H(K) \leftarrow iH(K_1) \leftarrow iH(K_2) \leftarrow iH(K_3) \leftarrow 0,$

$A_3 = \operatorname{Im}(i: A_2 \to A_2)$

$= H(K) \leftarrow iH(K_1) \leftarrow iiH(K_2) \leftarrow iiH(K_3) \leftarrow 0,$

and in A_4, all maps are inclusions. In particular, considering the exact couple

$$\begin{array}{ccc} A_4 & \to & A_4 \\ & {}_{k_4}\nwarrow \quad \swarrow_{j_4} & \\ & B_4 & \end{array}$$

we get $k_4=0$, and hence $\partial=0$ and $H(B_4)=B_4$. We write

$A_4 = A_\infty$, $B_4 = B_\infty$.

Note that A_∞ is the direct sum of the groups

$\dots = H(K) = H(K) \supseteq iH(K_1) \supseteq iiH(K_2) \supseteq iiiH(K_3) \supseteq 0.$

Since B_∞ is the quotient of A_∞ by i_∞, it is the associated graded complex.

The general case is similar: The sequence of subcomplexes

$$\dots = K = K \supset K_1 \supset K_2 \supset \dots$$

induces a sequence in cohomology

$$\dots \to H(K) \xleftarrow{i} H(K) \xleftarrow{i} H(K_1) \xleftarrow{i} H(K_2) \leftarrow \dots$$

and hence, by taking images in $H(K)$, a filtration

$$H(K) = F_0 \supset F_1 \supset F_2 \supset \dots.$$

The example shows the following: If the length of the filtration of K is finite then A_r, B_r are eventually stationary and

$$B_\infty = \bigoplus F_p/F_{p+1}$$

is the associated graded complex of $H(K)$.

Notation: $E_1 = H(B)$ with differential $d_1 = j_1 \circ k_1$,

$E_2 = H(E_1)$ with differential $d_2 = j_2 \circ k_2$,

etc.

We say that the spectral sequence (E_r, d_r) converges to a filtered group H if $E_\infty = \operatorname{Gr} H$.

Now let K be a filtered complex with a grading $K = \bigoplus_{n\in\mathbb{Z}} K^n$ (dimension). The filtration $\{K_p\}$ induces a filtration $K_p^n = K^n \cap K_p$ in each dimension. The argument before proves the following:

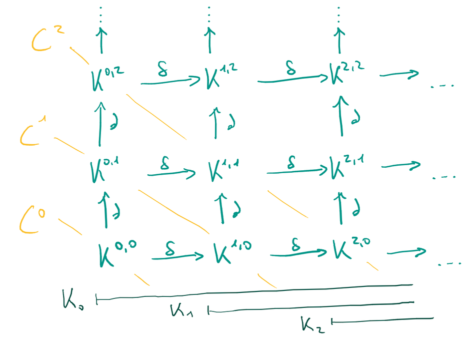
Note: $H(K)$ is graded complex. Hence, by definition, the E_r spectral sequence increases dimension by 1.

Let K be a graded filtered complex, with filtered cohomology $H_D^*(K)$. Assume the filtration induced in each dimension has finite length. The the s.e.s.

$$0 \to \bigoplus K_{p+1} \to \bigoplus K_p \to \bigoplus K_p/K_{p+1} \to 0$$

induces a spectral sequence which converges to $H_D^*(K)$.

Now let $K = \bigoplus K^{p,q}$ be a double complex.



We form a single complex $K = \bigoplus C^k$, $C^k = \bigoplus_{p+q=k} K^{p,q}$, with differential

$$D: C^k \to C^{k+1}, \quad D = \delta + (-1)^p \partial.$$

We define a filtration

$$K = K_0 \supset K_1 \supset K_2 \supset \dots, \quad K_p = \bigoplus_{i \geq p} \bigoplus_q K^{i,q}.$$

Then K is a graded filtered complex.

The direct sum

$$A = \bigoplus K_p$$

(a trivial double complex, with differentials $K_p \to K_p$)

also a double complex, from which we again form the single complex

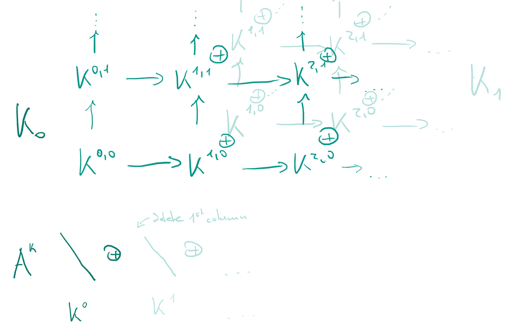
$$A = \bigoplus A^k.$$

We get an inclusion $i: A^k \to A^k$,

$$A^k \cap K_{p+1} \to A^k \cap K_p.$$

A inherits the differential operator

$$D = \delta + (-1)^p \partial.$$



The inclusion $i: A^k \to A^k$ is $A^k \cap F_{p+1} \to A^k \cap F_p$:

$A^1 = \dots \oplus K^{0,1} \oplus K^{1,0} \oplus \dots$

$A^2 = \dots \oplus K^{0,2} \oplus K^{1,1} \oplus K^{2,0} \oplus \dots$

Now, $B = A/i(A)$ is as a vector space the single complex formed from $K = \bigoplus K^{p,q}$, but its differential D is just $(-1)^p\partial$, since $\delta: K_p \to K_{p+1}$. We get a s.e.s. of graded complexes

$$0 \to A \to A \to B \to 0,$$

which induces a spectral sequence with

$$E_1 = H_D(B) = H_\partial(K).$$

What is E_2? The map $k_1: H(B) \to H(A)$ is constructed from

$$0 \to A^{k+1} \cap K_{p+1} \xrightarrow{i} A^{k+1} \cap K_p \xrightarrow{j} K^{p,k+1-p} \to 0$$

$$0 \to A^k \cap K_{p+1} \to A^k \cap K_p \to K^{p,k-p} \to 0.$$

We start with some $b \in A^k \cap K_p$ representing some $[b] \in K^{p,k-p}$ with $\partial b = 0$. Then we calculate $Db = \delta b + (-1)^p \partial b = \delta b$, and set

$$k_1[b] = [i^{-1}\delta b]_{H(A)}$$

Hence, $j_1 k_1 [b] = [\delta b]_{H_\partial}$

Hence, the differential on $E_1 = H(B)$ is given by $\delta = j_1 k_1$, and so,

$$E_2 = H_\delta E_1 = H_\delta H_\partial(K).$$

What is E_3? An element of $E_2 = H_\delta H_\partial(K)$ is represented by an element $b \in K$ such that

$$\partial b = 0,$$

$$\delta b = -(-1)^p \partial c \text{ for some } c \in K.$$

Let the class of b in E_r be denoted by $[b]_r$. Then

$$d_2[b]_2 = j_2 k_2 [b]_2 = j_2 k_1 [b]_1.$$

To compute $j_2 k_1 [b]_1$, we need to find $a \in K$ such that $k_1[b]_1 = i_1[a]_1$, so that $j_2 k_1 [b]_1 = [j_1 a]_2$.

Now, $k_1 b \in A^{k+1} \cap K_{p+1}$, so $a \in A^{k+1} \cap K_{p+2}$. We have $[b]_2 = [b+c]_2$, and

$$k_1(b+c) = D(b+c) = \delta b + (-1)^p \partial b + \delta c + (-1)^p \partial c = \delta c$$

Hence, $d_2[b_2] = [\delta c]_2$.

What does it mean to be in E_3? If $d_2[b]_2 = 0$, i.e. $[b]_2$ represents a cohomology class in E_3, then

- $\partial b = 0$, $\delta b = -(-1)^p \partial c_1$
- $c_1 = -(-1)^p \partial c_2$

Similarly, $b \in E_r$ represents a cohomology class in E_r iff it can be extended to a zig-zag of length r, with differential

$$d_r[b]_r = [\delta c_{r-1}]_r.$$

Summarising, we see that

$$E_r = \bigoplus E_r^{p,q}, \quad d_r : E_r^{p,q} \to E_r^{p+r, q-r+1}.$$

Furthermore, the filtration

$$H(K) = F_0 \supset F_1 \supset \dots$$

induces filtrations

$$H^n(K) = (F_0 \cap H^n) \supset (F_1 \cap H^n) \supset \dots \supset (F_n \cap H^n) \supset 0.$$

Hence,

Thm: There is a spectral sequence $\{E_r, d_r\}$ converging to $H_D(K)$ such that each E_r is bigraded, $d_r: E_r^{p,q} \to E_r^{p+r,q-r+1}$ and

$$GH_D^n(K) = \bigoplus_{p+q=n} E_\infty^{p,q}(K).$$

Aside: For two open sets U, V, we have a s.e.s.

$$0 \to \Omega^*(U\cup V) \to \Omega^*(U) \oplus \Omega^*(V) \xrightarrow{\delta} \Omega^*(U\cap V) \to 0$$

We will use this to get a spectral sequence. Consider the double complex

q			
2	$\Omega^2(U)\oplus\Omega^2(V)$	$\Omega^2(U\cap V)$	0 …
1	$\Omega^1(U)\oplus\Omega^1(V)$	$\Omega^1(U\cap V)$	0 …
0	$\Omega^0(U)\oplus\Omega^0(V)$	$\Omega^0(U\cap V)$	0 …
q/p	0	1	2

We claim that the associated single complex $C^*(\{U,V\},\Omega^*)$ computes the de Rham cohomology of $M = U\cup V$.

Let

$$r: \Omega^*(M) \to \Omega^*(U) \oplus \Omega^*(V)$$

denote the restriction. We have

$$D \circ r = (\delta + (-1)^p \partial) \circ r = \partial r,$$

so that r induces a map $H^*_{dR}(M) \to H^*(C^*(\{U,V\},\Omega^*))$.

We consider r as a map from the double complex to the double complex above.

We have $C^*(\{U,V\},\Omega^*) = K^{0,q} \oplus K^{1,q-1}$. If $\alpha = \alpha_0 + \alpha_1$ is a q-cochain then $\alpha_1 = \delta\beta$ for some $\beta \in K^{0,q}$, by Mayer-Vietoris. Then $\alpha, \alpha - D\beta$ are cohomologous. Hence, cochains are represented by $K^{0,q}$ cochains. For these representatives,

$$D\phi = 0 \quad \text{iff} \quad \partial\phi = \delta\phi = 0.$$

In particular, r is surjective.

For injectivity, assume that $r(\omega) = D\phi$. Write $\phi = \phi' + D\phi''$ for $\phi' \in K^{0,q}$. Then

$$r(\omega) = D\phi' = \partial\phi' \quad \text{and} \quad \delta\phi' = 0.$$

(hence, global form)

So ω is already exact.

Example: Let M be a manifold, $\mathcal{U}$ a good cover. Let

$$K^{p,q} = C^p(\mathcal{U}, \Omega^q) = \prod \Omega^q(U_{\alpha_0 \dots \alpha_p}).$$

We first consider the spectral sequence with the opposite filtration, so that $E_1 = H_\delta(K)$. Since the rows are exact, we get

$$E_1 = \begin{array}{ccc} \Omega^3(M) & 0 & 0 \dots \\ \Omega^2(M) & 0 & \dots \\ \Omega^1(M) & 0 & \dots \\ \Omega^0(M) & 0 & \dots \end{array} \Rightarrow E_2 = H_\partial H_\delta(K) = \begin{array}{cc} H^3_{dR}(M) & 0 \dots \\ H^2_{dR}(M) & 0 \dots \\ H^1_{dR}(M) & 0 \dots \\ H^0_{dR}(M) & 0 \dots \end{array}$$

So the spectral sequence degenerates at E_2. Hence,

$$H^n_{dR}(M) = \bigoplus_{p+q=n} H_D^n(C^*(\mathcal{U},\Omega^*)).$$

Taking the other filtration, since the columns are exact, we get

$$E_1 = H_\partial(K) = \begin{array}{ccc} 0 & 0 & 0 \\ C^0(\mathcal{U},\mathbb{R}) & C^1(\mathcal{U},\mathbb{R}) & C^2(\mathcal{U},\mathbb{R}) \dots \end{array}$$

$$\Rightarrow E_2 = H_\delta H_\partial(K) = \begin{array}{ccc} 0 & 0 & 0 \\ H^0(\mathcal{U},\mathbb{R}) & H^1(\mathcal{U},\mathbb{R}) & H^2(\mathcal{U},\mathbb{R}) \dots \end{array}$$

Hence,

$$H^*(\mathcal{U},\mathbb{R}) = H_D^*(C^*(\mathcal{U},\Omega^*)) = H^*_{dR}(M).$$

Example: Let $\pi: E \to M$ be a fibre bundle with fibre F, $\mathcal{U}$ a good cover of M. Let

$$K^{p,q} = C^p(\pi^{-1}\mathcal{U}, \Omega^q).$$

The associated spectral sequence has

(presheaf $U \mapsto H^q(\pi^{-1}U)$)

$$E_1^{p,q} = H_\partial(K)^{p,q} = \prod H^q(\pi^{-1}U_{\alpha\dots}) = C^p(\mathcal{U}, \mathcal{H}^q(F))$$

Since $\mathcal{U}$ is a good cover, $\mathcal{H}^q$ is locally constant with fibre $H^q(F)$. We get

$$E_2^{p,q} = H^p_\delta(\mathcal{U}, \mathcal{H}^q(F)).$$

This spectral sequence converges to $H^*(E)$, by the example on the left. If M is simply connected (and $H^*(F)$ finite dimensional) then $\mathcal{H}^q(F)$ is the constant presheaf $\mathbb{R}^{\dim H^q(F)}$. Hence,

$$E_2^{p,q} = H^p(\mathcal{U}, \mathbb{R}^{\dim H^q(F)}) = H^p(\mathcal{U},\mathbb{R})^{\dim H^q(F)} = H^p(M)\otimes H^q(F).$$

In the case of the trivial fibration, elements of E_2 are represented by global forms on $M\times F$. Hence, δ vanishes identically then and $E_2 = E_\infty$. We get Künneth's formula

$$H^*(M\times F) = H^*(M)\otimes H^*(F).$$

This argument also proves the Leray–Hirsch theorem, which (instead of triviality) only requires that elements of E_2 are represented by global forms, i.e. that $H^*(E) \twoheadrightarrow H^*(F)$ $\forall x \in M$, where $i_x : F \to E_x \subseteq E$.

Application: Cohomology of $\mathbb{CP}^n$

We have a circle bundle $S^{2n+1} \to \mathbb{CP}^n$. Since $\mathbb{CP}^n$ has no 1-skeleton, it is simply connected, so $E_2^{p,q} = H^p(\mathbb{CP}^n)\otimes H^q(S^1)$.

Hence, E_2 only has two identical rows $H^*(\mathbb{CP}^n)$:

$$E_2 = \begin{array}{c|cccc} 1 & \mathbb{R} & X_1 & X_2 & \dots \\ 0 & \mathbb{R} & X_1 & X_2 & X_3 \dots \\ \hline & 0 & 1 & 2 & 3 \end{array}$$

Since $\mathbb{CP}^n$ has dimension $2n$, $H^p(\mathbb{CP}^n) = 0$ for $p > 2n$. Clearly, $d_4 = d_5 = \dots = 0$.

Hence, the spectral sequence degenerates at the E_3-term, so

$$E_3 = E_4 = \dots = E_\infty = H^*(S^{2n+1})$$

and

$$E_3^{p,q} = \begin{cases} \mathbb{R} & \text{if } p=q=0 \text{ or } p=2n, q=1 \\ 0 & \text{o/w} \end{cases}$$

$$E_3 = \begin{array}{cccc} 0 & 0 & \dots & \mathbb{R} \\ \mathbb{R} & 0 & \dots & 0 \\ 0 & 1 & \dots & 2n \end{array}$$

In particular, the maps on E_2 must be isomorphisms. Hence,

$$E_2 = \begin{array}{cccccccc} \mathbb{R} & 0 & \mathbb{R} & 0 & \mathbb{R} & \dots & \mathbb{R} \\ \mathbb{R} & 0 & \mathbb{R} & 0 & \mathbb{R} & \dots & \mathbb{R} \end{array} \leftarrow H^*(\mathbb{CP}^n)$$